

# MADHAVAN S

Electronics & Communication Engineer

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## 👤 SUMMARY

Electronics & Communication Engineering undergraduate at PSG iTech and IIT Madras with hands-on experience in RTL design and verification, digital hardware and EDA tools. Built an agentic AI framework for RTL verification and coverage gap, corner-case identification with delta analysis, alongside hardware development experience at DRDO. IEEE author and DVCon India Fellow 2026 with a strong interest in Embedded Systems, VLSI design and verification.

## 🎓 EDUCATION

**PSG Institute of Technology and Applied Research** 2023 – 2027 (expected)  
B.E. Electronics and Communication Engineering - CGPA: 8.22 (until 6th semester)

**Indian Institute of Technology, Madras** 2023 – 2027 (expected)  
B.S. Electronics Systems (Hybrid) - CGPA: 8.00 (until now)

**ARB International School** 2022 – 2023  
Higher Secondary (Grade 12) - Percentage: 95.2%

## 📍 AREAS OF INTEREST

◦ RTL Design & Verification ◦ Digital IC Design ◦ VLSI ◦ EDA ◦ Embedded Systems ◦ Signal Processing

## ✂ SKILLS

**HDL & Programming:** Verilog, C, Embedded C, Assembly, Python (basic)

**EDA:** Xilinx Vivado, Cadence Virtuoso

**Tools:** STM32CubeIDE, MATLAB, ThingSpeak

**Hardware:** STM32F407, Zedboard Zynq 7000 (FPGA), Raspberry Pi, ESP32

**Soft Skills:** Leadership, Organising, Communication, Problem Solving

## 📁 EXPERIENCE

**Hardware Research Intern** May 2026 – July 2026

DRDO - CVRDE, Avadi (On-site)

- Designed a SSPC hardware interface with STM32F407, isolated gate driver, power MOSFET, and high-side Kelvin shunt current sensing; interfaced the conditioned signal to the 12-bit ADC for real-time current telemetry.
- Designed dual path overcurrent protection combining software I<sup>2</sup>t monitored with hardware DESAT protection and validated the STM32F407 ADC interface using STM32CubeIDE.

**AI/IoT Intern** July 2025 – Jan 2026

Do365 Technologies, RS Puram, Coimbatore (Hybrid)

- Developed and deployed an AI-based attendance system on Raspberry Pi 5, integrating custom YOLOv8 person detection, MobileNetV2 verification, and anti-spoofing to detect photo/screen-based impersonation; achieved 96% detection accuracy.
- Developed an IoT-based automated attendance workflow for real-time attendance logging across company campus and mess facilities.

## 📖 PUBLICATIONS

[Real-Time Detection of AI-Generated Voice Using Machine Learning and MATLAB Simulink](#)

DOI: 10.1109/ICIIET65921.2025.11379238

IEEE ICIIET 2025

- Designed a real-time AI-generated voice detection pipeline in MATLAB Simulink, extracting 26 audio features using MFCC and spectral features from 25 ms windows with 10 ms hop.
- Trained a 60-tree Random Forest achieving 92.3% accuracy, 93.8% recall, and 180 ms CPU inference latency, outperforming CNN/LSTM models under deployment constraints.

[A Deep Learning Framework for Spatiotemporal Air Quality Forecasting Using Episode Aware Optimization](#)

Presented, yet to be listed in IEEE Xplore

IEEE IC(SEC)<sup>2</sup> 2026

- Developed a lightweight ConvLSTM2D encoder-decoder for 16-hour PM2.5 forecasting using WRF-Chem data, incorporating episode-aware weighting and residual skip connections.
- Achieved 0.946 Pearson correlation, 18.24  $\mu\text{g}/\text{m}^3$  RMSE, and 88% AQI classification accuracy with only 402K trainable parameters.

## ACHIEVEMENT

### **First Runner-up – SAP LABS Hackfest'26**

Prize: Rs. 75,000

Built an agentic AI system for VLSI design verification capable of autonomous verification and correction through to the fabrication-ready stage.

### **DVCon India Fellowship**

2026

Selected as a DVCon India Fellow for the Design and Verification Conference (DVCon) 2026, recognising research interest towards RTL verification and hardware design.

### **Young Innovator Award – PSG iTech'26**

2026

Received the award at the annual Daksha 2026 summit for leading innovative projects, securing top positions in hackathons and achieving strong academic performance.

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## PROJECTS

### **VIGIL AI – RTL Verification Impact and Coverage-Gap Framework** *VLSI, RTL Verification, Agentic AI*

Built an agentic AI-based RTL verification framework that compares RTL versions, identifies modified modules, analyses downstream impact and prioritises relevant test cases. Detects coverage gaps and corner cases, including boundary conditions, FSM transitions, and interface issues; analyses regression results and generates actionable verification reports.

### **Intelligent Grid Monitoring System**

*Embedded Systems, Signal Processing, AI*

Designed a 230 V AC signal-conditioning and monitoring system using a bridge rectifier, LPF and voltage divider for ESP32 ADC interfacing, with frequency measurement. Developed fault detection using temporal parameter monitoring and a DNN on Raspberry Pi for fault classification, with automatic relay-based load disconnect and restoration.

### **Soil Savvy**

*AI, IoT, Agriculture*

Built an ESP32-based IoT system integrating NPK, pH, and soil-moisture sensors, with cloud data processing. Developed an ML pipeline combining soil-type classification and crop recommendation using real-time and historical data, achieving 94% accuracy, with results presented through a web interface.

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## LEADERSHIP & OUTREACH

### **Hackathon Organizer – iTech Summit, YUKTA 2K26**

2026

PSG Institute of Technology and Applied Research

Organised a 3 round inter college hackathon with 500+ Round 1 participants, managing judging coordination and participant communication; received the Organizing Excellence Award 2026 from the Principal.

### **Student Mentor**

February 2026

DMI Engineering College, Nagercoil

Invited as a guest speaker for the Student Mentorship Programme; delivered a career guidance session on hackathons, competitions, academics, and career development.

### **Industrial Visit Coordinator**

2024 & 2025

PSG Institute of Technology and Applied Research

Organised and managed industrial visits for 100+ students across two consecutive years, coordinating travel, accommodation, itineraries and end to end logistics.

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